

Project Report on
Crop Yield Prediction using Machine Learning

For

CSUT 307 Research Project-I

Submitted By

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For



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Index

Title	Page No
1. Introduction	
1.1 Existing System and Need for System	4 - 5
1.2 Scope of Work	5
1.3 Operating Environment – Hardware and Software	6
2. Proposed System	
2.1 Proposed System	7
2.1.1 Feasibility Study	
2.2 Objectives	7
2.3 Research Requirements	8
2.4 Literature Review	8 - 10
3. Analysis	
3.1 Exploratory Data Analysis	11
3.2 Dataset	11
4. Result and Discussion	
4.1 Research Methods	12
4.2 Experiment	12
4.3 Results	13
5. Annexure	
5.1 Annexure1: Input Screens	14
5.2 Annexure2: Output Report.	15 - 17
6. Future Enhancements	18
7. Conclusion	19
8. Bibliography	19

1. Introduction

1.1 Existing System

Crop yield prediction has traditionally been approached through **statistical models** and **ratio-based calculations** that primarily rely on structured agricultural records, such as area under cultivation and total production. While these approaches provide baseline yield estimates, they are constrained by several critical limitations.

1. Firstly, the **scope of features** in existing systems is restricted. Traditional models often rely only on historical tabular data, overlooking other influential parameters such as soil fertility, climatic variability, and remote sensing indices. As highlighted by **Bhat et al. (2024)**, yield is significantly shaped by factors like fertilizer inputs, soil quality, temperature, and solar radiation, which are rarely accounted for in conventional methods.
2. Secondly, the **predictive capacity** of traditional methods is limited. Classical regression-based systems tend to oversimplify the non-linear relationships among environmental, seasonal, and management factors. **Jabed and Murad (2024)** note that while early methods relied on expert observations and linear statistical models, they often failed to generalize across diverse geographies and crop varieties.
3. Thirdly, **data inconsistencies and quality issues** remain a persistent challenge. Agricultural datasets, such as those compiled at district and crop levels, often contain missing values, imbalanced crop records, or inconsistencies across years. **Panigrahi et al. (2023)** demonstrated that yield prediction models trained on such heterogeneous datasets require extensive preprocessing and may still deliver suboptimal accuracy.
4. Moreover, conventional systems typically provide **static insights**—annual yield statistics aggregated at district or state levels. While informative for long-term planning, such outputs lack **real-time or dynamic forecasting capabilities** essential for in-season decision-making. Studies in **India (Jhajharia et al., 2023; Vinson et al., 2022)** have shown that incorporating dynamic weather, soil, and management parameters into ML frameworks significantly improves yield forecasting.
5. Finally, existing frameworks display a **lack of integration and scalability**. They are rarely designed to assimilate multi-source data (e.g., climate models, soil databases, satellite imagery), which limits their adaptability across diverse crops, seasons, and agro-climatic zones. **Rajakumaran et al. (2024)** emphasize that advanced hybrid ML methods, such as weighted tree-based SVMs with feature selection, outperform traditional techniques in handling such heterogeneity.

Need for System

The proposed system integrates R Studio, Power BI, and Python to address the following challenges in the existing system:

1. **Comprehensive Analysis:** By using multiple platforms, we combine the statistical strengths of R Studio, the visualization capabilities of Power BI, and the versatility of Python.
2. **Improved Predictions:** Advanced machine learning models are implemented in R and Python, ensuring accuracy and reliability in predictions.
3. **Interactive Insights:** Power BI's dashboards allow stakeholders to explore data and predictions interactively, making the results accessible and actionable.
4. **Cross-validation:** Implementing the project across three platforms provides a robust mechanism to cross-validate results, ensuring model reliability and reproducibility.

1.2 Scope of Work

1. **Data Preprocessing:**
 - **Python:** Used for cleaning, imputing missing values, encoding categorical data, and scaling features.
 - **R Studio:** Performed additional preprocessing tasks to verify consistency and suitability for statistical modeling.
2. **Exploratory Data Analysis (EDA):**
 - **Power BI:** Created dynamic dashboards and interactive visualizations to explore relationships between features.
 - **R Studio:** Conducted correlation analysis and visualized distributions using ggplot2.
3. **Model Development:**
 - Trained five machine learning models in **Python** (Linear Regression, Random Forest, KNN, Decision Tree, and SVM).
 - Evaluated model accuracy and compared results with Python implementations.
4. **Visualization and Reporting:**
 - Visualized feature importance and model evaluation metrics in R Studio.
 - Integrated Python outputs into **Power BI** dashboards for a unified presentation.
5. **Final Comparison:**
 - Compiled a comprehensive report detailing insights and results.

1.3 Operating Environment – Hardware and Software

Hardware Requirements:

- **Processor:** Intel Core i5 or i7
- **RAM:** Minimum 8 GB (recommended for handling large datasets)
- **Storage:** Minimum 50 GB of free disk space

Software Requirements:

1. R Studio:

- Version: 2023.06 or later
- Libraries: caret, ggplot2, random Forest, e1071, FNN, dplyr

2. Python:

- Version: Python 3.10 or later
- Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn

3. Power BI:

- Version: Power BI Desktop 2.138.782.0

4. Operating System:

- Windows 10/11 (64-bit)

2. Proposed System

The proposed system is designed to address the limitations of the existing system by leveraging a robust workflow for predicting house prices. It integrates advanced machine learning techniques, comprehensive exploratory data analysis, and interactive visualizations to deliver accurate predictions and actionable insights.

The system focuses on the following key aspects:

1. **Data Preprocessing:** Ensuring the dataset is clean, consistent, and ready for analysis by handling missing values, encoding categorical variables, and scaling numerical features.
2. **Model Evaluation:** Implementing and comparing multiple machine learning algorithms to identify the best-performing model.
3. **Visualizations:** Using dynamic visualizations to better understand feature relationships and prediction results.
4. **Optimization:** Employing hyperparameter tuning to refine the performance of the selected model for greater accuracy.

2.1.1. Feasibility Study

The feasibility study analyzes the practicality of implementing the proposed system in terms of the following aspects:

1. **Technical Feasibility:**
 - The system uses readily available libraries and tools for machine learning, such as R and Python, and visualization software like Power BI.
 - Hardware requirements are standard, and no specialized infrastructure is necessary.
2. **Economic Feasibility:**
 - Open-source tools are primarily utilized, minimizing costs.
 - Training and implementation costs are justified by the system's potential to improve decision-making in yield Prediction.
3. **Operational Feasibility:**
 - The system is user-friendly and designed to provide insights that are easily interpretable by stakeholders, such as farmers and pesticides shopkeepers.
 - Integration of predictive results into dynamic dashboards ensures accessibility and usability.

2.2 Objectives

The objectives of the proposed system are as follows:

1. To accurately predict Crop Yield based on various features (Area, Production, Season) provided in the dataset.
2. To Check the performance of ML models based on different Scenario of Datasets.
3. To identify key factors influencing Yield through data analysis and visualization.
4. To evaluate and compare the performance of multiple machine learning models, including Linear Regression, Random Forest, K-Nearest Neighbors, and Decision Tree.
5. To optimize the selected model for maximum predictive accuracy.
6. To present the findings in an interactive and user-friendly format.

2.3 Research Requirements

1. **Dataset:** The dataset ([Maha_Agri.csv](#)) sourced from Kaggle, containing features like crop, season, area, production, and the target variable [yield](#).
2. **Tools:**
 - Statistical and machine learning tools for analysis and modeling.
 - Visualization tools for dynamic and static representation of results.
3. **Knowledge Base:**
 - Proficiency in machine learning algorithms.
 - Understanding of statistical analysis and data preprocessing techniques.
4. **Metrics:** Use of Mean Squared Error (MSE) and R-squared (R^2) to evaluate model performance.

2.4 Literature Review

Sr. No.	Paper Name	Authors	Review
1	Crop Yield Prediction using WTDCNN	Subramaniam & Marimuthu (2024)	Proposed WTDCNN with modified Whale Optimization Algorithm, achieving 98.96% accuracy. Integrated meteorological and agricultural data for high-precision yield prediction.
2	Crop Recommendation using ML	Surana & Khandelwal (2024)	Used Random Forest for crop recommendation with 99.93% accuracy. Considered soil pH, rainfall, and temperature to suggest suitable crops.

3	Maize Yield Prediction with XGBoost	Bhat et al. (2024)	Employed XGBoost with Grid Search for maize yield prediction. Used Explainable AI to analyze factors like vapor pressure and soil nutrients.
4	Hybrid CLSTMT Model for Crop Yield	Pravesh et al. (2024)	Introduced hybrid CLSTMT (Clustering + LSTM + Transformer) achieving $R^2 = 0.951$. Captured climate-influenced yield trends using global data (1990–2013).
5	AI Techniques for Crop Yield Prediction	Jabed & Murad (2024)	Conducted SLR identifying top AI models (RF, ANN, CNN, LSTM). Highlighted need for quality data (NDVI, EVI, LAI) and integration of AI in agriculture.
6	ML and DL Approaches in Yield Prediction	Shawon et al. (2025)	Reviewed 97 papers using PRISMA. Found RF, GBT, CNN, and LSTM as dominant models. Discussed key features and evaluation metrics (RMSE, MAE, R^2).
7	Crop Yield Prediction in Rajasthan	Jhajharia et al. (2023)	Compared ML/DL models; Random Forest achieved best $R^2 = 0.963$. Provided practical insights for crop selection and productivity improvement.
8	ML Regression Models for Yield	Panigrahi et al. (2023)	Built six ML regressors; Random Forest showed best MAE = 468.16. Emphasized precision agriculture and data-driven farm decisions.
9	Paddy Yield Prediction using GRNN	Joshua et al. (2022)	Used GRNN on 18 years of Tamil Nadu data; achieved 97% accuracy. Demonstrated NN effectiveness for nonlinear agricultural dependencies.
10	MAWT-SVM for Crop Yield Prediction	Rajakumaran et al. (2024)	Proposed hybrid MAWT-SVM (using PCA & GA), achieving 98.5% accuracy. Outperformed CNN and DNN in multi-attribute yield estimation.

3. Analysis

3.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the dataset's structure, identify patterns, and uncover relationships between features and the target variable (Yield). Key steps included:

1. Understanding the Dataset:

- Summary statistics (mean, median, range) were computed for all numerical features to understand their distributions.
- Categorical features were analyzed for unique values and frequency counts.

2. Correlation Analysis:

- A correlation matrix was created to identify features with strong linear relationships with yield.
- Highly correlated features such as area and production showed a strong positive relationship with yield.

3. Visualizations:

- Histograms: Histograms were plotted for features like Area and Yield, Crop and Yield to understand their distribution and detect skewness or outliers.

4. Insights from EDA:

- The data is Somewhat left-skewed, some crops have high production unit (tons) than others like sugarcane, maize etc.
- There are some fluctuations in data with respect to Season (Kharif, Rabi and Whole Year).
- There are some outliers in data w.r.t Yield per Crop.

3.2 Dataset

The dataset has been sourced from Kaggle (Maha_Agri.csv). It consists of the following attributes:

1. Numerical Features:

- **Yield:** The target variable, representing the Yield per hectare.
- **Area:** Total area of the production of that crop.
- **Production:** Tons of productions of a crop w.r.t total area.

2. Categorical Features:

- **District:** Names of districts from Maharashtra.
- **Crop:** Crop Names from that District
- **Season:** Season Name (Kharif, Rabi and Whole Year).
- **Year:** Year Data (2010 to 2020).

3. Preprocessing Steps:

- Missing Values: Imputed using median values for numerical features and mode for categorical features.
- Encoding: Categorical variables were one-hot encoded to convert them into numerical format.
- Scaling: Numerical features were scaled to standardize their values.

The dataset was split into an 80-20 ratio for training and testing purposes, ensuring the target variable's distribution remained consistent across both subsets.

4. Result and Discussion

4.1 Research Methods

To predict Yield accurately, multiple machine learning algorithms with different scenarios of dataset were implemented and evaluated. The research methodology followed a structured approach:

1. Preprocessing:

- Missing values were handled appropriately, i.e. (Outliers removed and outliers capped).
- Categorical variables were encoded using one-hot encoding.
- Numerical features were standardized to improve model performance.

2. Model Selection:

Five machine learning models were chosen for evaluation:

- **Linear Regression:** Assumes a linear relationship between features and the target variable.
- **Random Forest:** An ensemble model that builds multiple decision trees to improve accuracy.
- **K-Nearest Neighbors (KNN):** A non-parametric algorithm based on the proximity of data points.
- **Decision Tree:** Splits data into branches for predictions.
- **AdaBoost:** An ensemble method that combines multiple weak learners (often decision trees) by iteratively adjusting weights to minimize errors.
- **XGBRegressor:** An optimized gradient boosting algorithm that builds trees sequentially, focusing on correcting errors from previous models; known for high performance and scalability.

3. Evaluation Metrics:

The models were evaluated using the following metrics:

- R-squared (R^2): Measures the proportion of variance explained by the model.
- Mean Squared Error (MSE): Calculates the average squared difference between predicted and actual values.

4.2 Experiment

The experiment involved the following steps:

1. Dataset Preparation:

The Dataset was modified in different categories, Data Without filtration, with filtration but Outliers present, removed outliers, capped outliers and finally removed some of the values from area, production and yield column.

2. Model Training and Evaluation:

Each model was trained on the training dataset and predictions were made on the test dataset.

3. Visualization of Results:

Model Comparisons: Bar charts were created to compare models and their R^2 values across all dataset categories.

4.3 Results

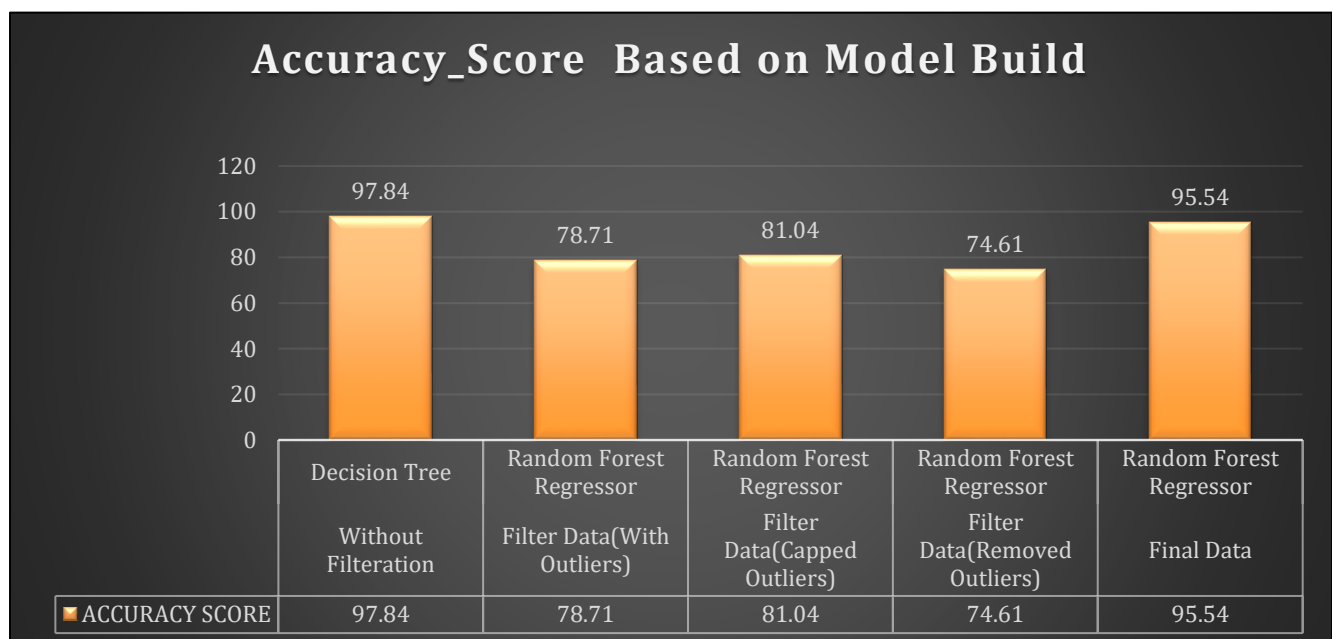


Fig -1: All Models Evaluation (Accuracy %)

1. **Decision Tree without filtration** achieved the highest accuracy (**97.84%**), showing that it can capture dataset patterns effectively without preprocessing.
2. **Random Forest Regressor with outlier handling** showed varied performance:
 3. With outliers: **78.71%**
 4. Capped outliers: **81.04%**
 5. Removed outliers: **74.61%**

This indicates that outlier treatment significantly impacts model performance, with capping being more effective than outright removal.

6. **Random Forest with final cleaned data** delivered strong performance (**95.54%**), close to the Decision Tree, demonstrating the effectiveness of preprocessing and feature refinement.
7. Overall, **ensemble methods (Random Forest)** proved more stable across different data treatments, whereas the **Decision Tree** performed best only without preprocessing.
8. The results highlight the importance of **data quality and preprocessing** in achieving reliable predictions, where handling outliers correctly leads to improved model robustness.

5. Annexure

5.1 Input Screens

Python:

```
In [54]: 1 import pandas as pd
2 import numpy as np
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 from sklearn.preprocessing import StandardScaler,MinMaxScaler
6 import warnings
7 warnings.filterwarnings('ignore')
8 from sklearn.model_selection import train_test_split
9 from sklearn.linear_model import LinearRegression
10 from sklearn.neighbors import KNeighborsRegressor
11 from sklearn.ensemble import RandomForestRegressor,AdaBoostRegressor,GradientBoostingRegressor
12 from sklearn.svm import SVC
13 from sklearn.model_selection import RandomizedSearchCV
14 from xgboost import XGBRegressor
15 from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
16 from sklearn.metrics import r2_score
17 plt.rcParams['figure.figsize']=(10,5)
18 plt.rcParams['figure.dpi']=200
19
20 In [56]: 1 # 1. DATA LOADING & INSPECTION
2 # Read the data
3 df = pd.read_excel("agri_haha(Reference).xlsx")
4
5 # Initial Inspection
6
7 print("First 5 Rows:")
8 print(df.head())
9 print("Last 5 Rows:")
10 print(df.tail())
11
```

```
first 5 Rows:
District  Crop  Year  Season  Area(ha)  Production(tons) \
0  ANURAGCHAD  Bajra  2008-11  Kharif  89800  120500
1  BEED  Bajra  2008-11  Kharif  122000  112000
2  DARE  Bajra  2008-11  Kharif  109000  171500
3  JALNAR  Bajra  2008-11  Kharif  35000  49000
4  NASHIK  Bajra  2008-11  Kharif  171200  230300

Last 5 Rows:
District  Crop  Year  Season  Area(ha)  Production(tons) \
1297  BOLHADA  Urad  2010-20  Kharif  10300  13000
1298  BOLHADA  Urad  2010-20  Kharif  1871  5564
1299  LATUR  Urad  2010-20  Kharif  7530  4207
1300  GONDIA  Wheat  2010-20  Rabi  3013  5008
1301  KOLHAPUR  Wheat  2010-20  Rabi  1300  8000
```

```
In [57]: 1 print("data types:")
2 print(df.dtypes)
3 print("data shape:")
4 print(df.shape)
5 print("data shape:")
6 print(df.shape)
7 print("missing values:")
8 print(df.isnull().sum())
9 print(df.isnull().sum())

data types:
District      object
Crop          object
Year          object
Season        object
Area(ha)      int64
Production(tons) int64
Yield(ton/ha) float64
dtype: object

Dataset Shape:
(1302, 7)

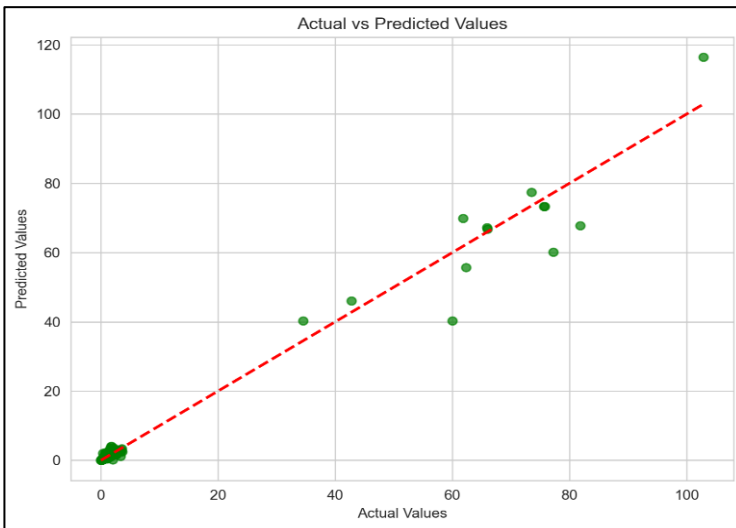
Missing values:
District      0
Crop          0
Year          0
Season        0
Area(ha)      0
Production(tons) 0
Yield(ton/ha) 0
```

```
In [58]: 1 # 2. DATA CLEANING
2 # Drop the rows with missing values
3 df = df.dropna()
4
5 In [59]: 1 data_ha = df[['Season']] = 'Kharif'.sort_values(ascending=True)
2 data_rabi = df[['Season']] = 'Rabi'.sort_values(ascending=True)
3 data_year = df[['Season']] = 'Whole Year'.sort_values(ascending=True)
4
5 In [60]: 1 data_ha.head()
6
7 Out[60]:
District  Crop  Year  Season  Area(ha)  Production(tons)  Yield(ton/ha)
0  ANURAGCHAD  Bajra  2008-11  Kharif  89800  120500  1.34
1  BEED  Bajra  2008-11  Kharif  122000  112000  0.92
2  DARE  Bajra  2008-11  Kharif  109000  171500  1.57
3  JALNAR  Bajra  2008-11  Kharif  35000  49000  1.40
4  NASHIK  Bajra  2008-11  Kharif  171200  230300  1.35
5  ANURAGCHAD  Bajra  2009-10  Kharif  89800  120500  1.34
6  BEED  Bajra  2009-10  Kharif  122000  112000  0.92
7  DARE  Bajra  2009-10  Kharif  109000  171500  1.57
8  JALNAR  Bajra  2009-10  Kharif  35000  49000  1.40
9  NASHIK  Bajra  2009-10  Kharif  171200  230300  1.35
10  ANURAGCHAD  Bajra  2010-11  Kharif  89800  120500  1.34
11  BEED  Bajra  2010-11  Kharif  122000  112000  0.92
12  DARE  Bajra  2010-11  Kharif  109000  171500  1.57
13  JALNAR  Bajra  2010-11  Kharif  35000  49000  1.40
14  NASHIK  Bajra  2010-11  Kharif  171200  230300  1.35
15  ANURAGCHAD  Bajra  2011-12  Kharif  89800  120500  1.34
16  BEED  Bajra  2011-12  Kharif  122000  112000  0.92
17  DARE  Bajra  2011-12  Kharif  109000  171500  1.57
18  JALNAR  Bajra  2011-12  Kharif  35000  49000  1.40
19  NASHIK  Bajra  2011-12  Kharif  171200  230300  1.35
20  ANURAGCHAD  Bajra  2012-13  Kharif  89800  120500  1.34
21  BEED  Bajra  2012-13  Kharif  122000  112000  0.92
22  DARE  Bajra  2012-13  Kharif  109000  171500  1.57
23  JALNAR  Bajra  2012-13  Kharif  35000  49000  1.40
24  NASHIK  Bajra  2012-13  Kharif  171200  230300  1.35
25  ANURAGCHAD  Bajra  2013-14  Kharif  89800  120500  1.34
26  BEED  Bajra  2013-14  Kharif  122000  112000  0.92
27  DARE  Bajra  2013-14  Kharif  109000  171500  1.57
28  JALNAR  Bajra  2013-14  Kharif  35000  49000  1.40
29  NASHIK  Bajra  2013-14  Kharif  171200  230300  1.35
30  ANURAGCHAD  Bajra  2014-15  Kharif  89800  120500  1.34
31  BEED  Bajra  2014-15  Kharif  122000  112000  0.92
32  DARE  Bajra  2014-15  Kharif  109000  171500  1.57
33  JALNAR  Bajra  2014-15  Kharif  35000  49000  1.40
34  NASHIK  Bajra  2014-15  Kharif  171200  230300  1.35
35  ANURAGCHAD  Bajra  2015-16  Kharif  89800  120500  1.34
36  BEED  Bajra  2015-16  Kharif  122000  112000  0.92
37  DARE  Bajra  2015-16  Kharif  109000  171500  1.57
38  JALNAR  Bajra  2015-16  Kharif  35000  49000  1.40
39  NASHIK  Bajra  2015-16  Kharif  171200  230300  1.35
40  ANURAGCHAD  Bajra  2016-17  Kharif  89800  120500  1.34
41  BEED  Bajra  2016-17  Kharif  122000  112000  0.92
42  DARE  Bajra  2016-17  Kharif  109000  171500  1.57
43  JALNAR  Bajra  2016-17  Kharif  35000  49000  1.40
44  NASHIK  Bajra  2016-17  Kharif  171200  230300  1.35
45  ANURAGCHAD  Bajra  2017-18  Kharif  89800  120500  1.34
46  BEED  Bajra  2017-18  Kharif  122000  112000  0.92
47  DARE  Bajra  2017-18  Kharif  109000  171500  1.57
48  JALNAR  Bajra  2017-18  Kharif  35000  49000  1.40
49  NASHIK  Bajra  2017-18  Kharif  171200  230300  1.35
50  ANURAGCHAD  Bajra  2018-19  Kharif  89800  120500  1.34
51  BEED  Bajra  2018-19  Kharif  122000  112000  0.92
52  DARE  Bajra  2018-19  Kharif  109000  171500  1.57
53  JALNAR  Bajra  2018-19  Kharif  35000  49000  1.40
54  NASHIK  Bajra  2018-19  Kharif  171200  230300  1.35
55  ANURAGCHAD  Bajra  2019-20  Kharif  89800  120500  1.34
56  BEED  Bajra  2019-20  Kharif  122000  112000  0.92
57  DARE  Bajra  2019-20  Kharif  109000  171500  1.57
58  JALNAR  Bajra  2019-20  Kharif  35000  49000  1.40
59  NASHIK  Bajra  2019-20  Kharif  171200  230300  1.35
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64  NASHIK  Bajra  2020-21  Kharif  171200  230300  1.35
65  ANURAGCHAD  Bajra  2021-22  Kharif  89800  120500  1.34
66  BEED  Bajra  2021-22  Kharif  122000  112000  0.92
67  DARE  Bajra  2021-22  Kharif  109000  171500  1.57
68  JALNAR  Bajra  2021-22  Kharif  35000  49000  1.40
69  NASHIK  Bajra  2021-22  Kharif  171200  230300  1.35
70  ANURAGCHAD  Bajra  2022-23  Kharif  89800  120500  1.34
71  BEED  Bajra  2022-23  Kharif  122000  112000  0.92
72  DARE  Bajra  2022-23  Kharif  109000  171500  1.57
73  JALNAR  Bajra  2022-23  Kharif  35000  49000  1.40
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84  NASHIK  Bajra  2024-25  Kharif  171200  230300  1.35
85  ANURAGCHAD  Bajra  2025-26  Kharif  89800  120500  1.34
86  BEED  Bajra  2025-26  Kharif  122000  112000  0.92
87  DARE  Bajra  2025-26  Kharif  109000  171500  1.57
88  JALNAR  Bajra  2025-26  Kharif  35000  49000  1.40
89  NASHIK  Bajra  2025-26  Kharif  171200  230300  1.35
90  ANURAGCHAD  Bajra  2026-27  Kharif  89800  120500  1.34
91  BEED  Bajra  2026-27  Kharif  122000  112000  0.92
92  DARE  Bajra  2026-27  Kharif  109000  171500  1.57
93  JALNAR  Bajra  2026-27  Kharif  35000  49000  1.40
94  NASHIK  Bajra  2026-27  Kharif  171200  230300  1.35
95  ANURAGCHAD  Bajra  2027-28  Kharif  89800  120500  1.34
96  BEED  Bajra  2027-28  Kharif  122000  112000  0.92
97  DARE  Bajra  2027-28  Kharif  109000  171500  1.57
98  JALNAR  Bajra  2027-28  Kharif  35000  49000  1.40
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101  BEED  Bajra  2028-29  Kharif  122000  112000  0.92
102  DARE  Bajra  2028-29  Kharif  109000  171500  1.57
103  JALNAR  Bajra  2028-29  Kharif  35000  49000  1.40
104  NASHIK  Bajra  2028-29  Kharif  171200  230300  1.35
105  ANURAGCHAD  Bajra  2029-30  Kharif  89800  120500  1.34
106  BEED  Bajra  2029-30  Kharif  122000  112000  0.92
107  DARE  Bajra  2029-30  Kharif  109000  171500  1.57
108  JALNAR  Bajra  2029-30  Kharif  35000  49000  1.40
109  NASHIK  Bajra  2029-30  Kharif  171200  230300  1.35
110  ANURAGCHAD  Bajra  2030-31  Kharif  89800  120500  1.34
111  BEED  Bajra  2030-31  Kharif  122000  112000  0.92
112  DARE  Bajra  2030-31  Kharif  109000  171500  1.57
113  JALNAR  Bajra  2030-31  Kharif  35000  49000  1.40
114  NASHIK  Bajra  2030-31  Kharif  171200  230300  1.35
115  ANURAGCHAD  Bajra  2031-32  Kharif  89800  120500  1.34
116  BEED  Bajra  2031-32  Kharif  122000  112000  0.92
117  DARE  Bajra  2031-32  Kharif  109000  171500  1.57
118  JALNAR  Bajra  2031-32  Kharif  35000  49000  1.40
119  NASHIK  Bajra  2031-32  Kharif  171200  230300  1.35
120  ANURAGCHAD  Bajra  2032-33  Kharif  89800  120500  1.34
121  BEED  Bajra  2032-33  Kharif  122000  112000  0.92
122  DARE  Bajra  2032-33  Kharif  109000  171500  1.57
123  JALNAR  Bajra  2032-33  Kharif  35000  49000  1.40
124  NASHIK  Bajra  2032-33  Kharif  171200  230300  1.35
125  ANURAGCHAD  Bajra  2033-34  Kharif  89800  120500  1.34
126  BEED  Bajra  2033-34  Kharif  122000  112000  0.92
127  DARE  Bajra  2033-34  Kharif  109000  171500  1.57
128  JALNAR  Bajra  2033-34  Kharif  35000  49000  1.40
129  NASHIK  Bajra  2033-34  Kharif  171200  230300  1.35
130  ANURAGCHAD  Bajra  2034-35  Kharif  89800  120500  1.34
131  BEED  Bajra  2034-35  Kharif  122000  112000  0.92
132  DARE  Bajra  2034-35  Kharif  109000  171500  1.57
133  JALNAR  Bajra  2034-35  Kharif  35000  49000  1.40
134  NASHIK  Bajra  2034-35  Kharif  171200  230300  1.35
135  ANURAGCHAD  Bajra  2035-36  Kharif  89800  120500  1.34
136  BEED  Bajra  2035-36  Kharif  122000  112000  0.92
137  DARE  Bajra  2035-36  Kharif  109000  171500  1.57
138  JALNAR  Bajra  2035-36  Kharif  35000  49000  1.40
139  NASHIK  Bajra  2035-36  Kharif  171200  230300  1.35
140  ANURAGCHAD  Bajra  2036-37  Kharif  89800  120500  1.34
141  BEED  Bajra  2036-37  Kharif  122000  112000  0.92
142  DARE  Bajra  2036-37  Kharif  109000  171500  1.57
143  JALNAR  Bajra  2036-37  Kharif  35000  49000  1.40
144  NASHIK  Bajra  2036-37  Kharif  171200  230300  1.35
145  ANURAGCHAD  Bajra  2037-38  Kharif  89800  120500  1.34
146  BEED  Bajra  2037-38  Kharif  122000  112000  0.92
147  DARE  Bajra  2037-38  Kharif  109000  171500  1.57
148  JALNAR  Bajra  2037-38  Kharif  35000  49000  1.40
149  NASHIK  Bajra  2037-38  Kharif  171200  230300  1.35
150  ANURAGCHAD  Bajra  2038-39  Kharif  89800  120500  1.34
151  BEED  Bajra  2038-39  Kharif  122000  112000  0.92
152  DARE  Bajra  2038-39  Kharif  109000  171500  1.57
153  JALNAR  Bajra  2038-39  Kharif  35000  49000  1.40
154  NASHIK  Bajra  2038-39  Kharif  171200  230300  1.35
155  ANURAGCHAD  Bajra  2039-40  Kharif  89800  120500  1.34
156  BEED  Bajra  2039-40  Kharif  122000  112000  0.92
157  DARE  Bajra  2039-40  Kharif  109000  171500  1.57
158  JALNAR  Bajra  2039-40  Kharif  35000  49000  1.40
159  NASHIK  Bajra  2039-40  Kharif  171200  230300  1.35
160  ANURAGCHAD  Bajra  2040-41  Kharif  89800  120500  1.34
161  BEED  Bajra  2040-41  Kharif  122000  112000  0.92
162  DARE  Bajra  2040-41  Kharif  109000  171500  1.57
163  JALNAR  Bajra  2040-41  Kharif  35000  49000  1.40
164  NASHIK  Bajra  2040-41  Kharif  171200  230300  1.35
165  ANURAGCHAD  Bajra  2041-42  Kharif  89800  120500  1.34
166  BEED  Bajra  2041-42  Kharif  122000  112000  0.92
167  DARE  Bajra  2041-42  Kharif  109000  171500  1.57
168  JALNAR  Bajra  2041-42  Kharif  35000  49000  1.40
169  NASHIK  Bajra  2041-42  Kharif  171200  230300  1.35
170  ANURAGCHAD  Bajra  2042-43  Kharif  89800  120500  1.34
171  BEED  Bajra  2042-43  Kharif  122000  112000  0.92
172  DARE  Bajra  2042-43  Kharif  109000  171500  1.57
173  JALNAR  Bajra  2042-43  Kharif  35000  49000  1.40
174  NASHIK  Bajra  2042-43  Kharif  171200  230300  1.35
175  ANURAGCHAD  Bajra  2043-44  Kharif  89800  120500  1.34
176  BEED  Bajra  2043-44  Kharif  122000  112000  0.92
177  DARE  Bajra  2043-44  Kharif  109000  171500  1.57
178  JALNAR  Bajra  2043-44  Kharif  35000  49000  1.40
179  NASHIK  Bajra  2043-44  Kharif  171200  230300  1.35
180  ANURAGCHAD  Bajra  2044-45  Kharif  89800  120500  1.34
181  BEED  Bajra  2044-45  Kharif  122000  112000  0.92
182  DARE  Bajra  2044-45  Kharif  109000  171500  1.57
183  JALNAR  Bajra  2044-45  Kharif  35000  49000  1.40
184  NASHIK  Bajra  2044-45  Kharif  171200  230300  1.35
185  ANURAGCHAD  Bajra  2045-46  Kharif  89800  120500  1.34
186  BEED  Bajra  2045-46  Kharif  122000  112000  0.92
187  DARE  Bajra  2045-46  Kharif  109000  171500  1.57
188  JALNAR  Bajra  2045-46  Kharif  35000  49000  1.40
189  NASHIK  Bajra  2045-46  Kharif  171200  230300  1.35
190  ANURAGCHAD  Bajra  2046-47  Kharif  89800  120500  1.34
191  BEED  Bajra  2046-47  Kharif  122000  112000  0.92
192  DARE  Bajra  2046-47  Kharif  109000  171500  1.57
193  JALNAR  Bajra  2046-47  Kharif  35000  49000  1.40
194  NASHIK  Bajra  2046-47  Kharif  171200  230300  1.35
195  ANURAGCHAD  Bajra  2047-48  Kharif  89800  120500  1.34
196  BEED  Bajra  2047-48  Kharif  122000  112000  0.92
197  DARE  Bajra  2047-48  Kharif  109000  171500  1.57
198  JALNAR  Bajra  2047-48  Kharif  35000  49000  1.40
199  NASHIK  Bajra  2047-48  Kharif  171200  230300  1.35
200  ANURAGCHAD  Bajra  2048-49  Kharif  89800  120500  1.34
201  BEED  Bajra  2048-49  Kharif  122000  112000  0.92
202  DARE  Bajra  2048-49  Kharif  109000  171500  1.57
203  JALNAR  Bajra  2048-49  Kharif  35000  49000  1.40
204  NASHIK  Bajra  2048-49  Kharif  171200  230300  1.35
205  ANURAGCHAD  Bajra  2049-50  Kharif  89800  120500  1.34
206  BEED  Bajra  2049-50  Kharif  122000  112000  0.92
207  DARE  Bajra  2049-50  Kharif  109000  171500  1.57
208  JALNAR  Bajra  2049-50  Kharif  35000  49000  1.40
209  NASHIK  Bajra  2049-50  Kharif  171200  230300  1.35
210  ANURAGCHAD  Bajra  2050-51  Kharif  89800  120500  1.34
211  BEED  Bajra  2050-51  Kharif  122000  112000  0.92
212  DARE  Bajra  2050-51  Kharif  109000  171500  1.57
213  JALNAR  Bajra  2050-51  Kharif  35000  49000  1.40
214  NASHIK  Bajra  2050-51  Kharif  171200  230300  1.35
215  ANURAGCHAD  Bajra  2051-52  Kharif  89800  120500  1.34
216  BEED  Bajra  2051-52  Kharif  122000  112000  0.92
217  DARE  Bajra  2051-52  Kharif  109000  171500  1.57
218  JALNAR  Bajra  2051-52  Kharif  35000  49000  1.40
219  NASHIK  Bajra  2051-52  Kharif  171200  230300  1.35
220  ANURAGCHAD  Bajra  2052-53  Kharif  89800  120500  1.34
221  BEED  Bajra  2052-53  Kharif  122000  112000  0.92
222  DARE  Bajra  2052-53  Kharif  109000  171500  1.57
223  JALNAR  Bajra  2052-53  Kharif  35000  49000  1.40
224  NASHIK  Bajra  2052-53  Kharif  171200  230300  1.35
225  ANURAGCHAD  Bajra  2053-54  Kharif  89800  120500  1.34
226  BEED  Bajra  2053-54  Kharif  122000  112000  0.92
227  DARE  Bajra  2053-54  Kharif  109000  171500  1.57
228  JALNAR  Bajra  2053-54  Kharif  35000  49000  1.40
229  NASHIK  Bajra  2053-54  Kharif  171200  230300  1.35
230  ANURAGCHAD  Bajra  2054-55  Kharif  89800  120500  1.34
231  BEED  Bajra  2054-55  Kharif  122000  112000  0.92
232  DARE  Bajra  2054-55  Kharif  109000  171500  1.57
233  JALNAR  Bajra  2054-55  Kharif  35000  49000  1.40
234  NASHIK  Bajra  2054-55  Kharif  171200  230300  1.35
235  ANURAGCHAD  Bajra  2055-56  Kharif  89800  120500  1.34
236  BEED  Bajra  2055-56  Kharif  122000  112000  0.92
237  DARE  Bajra  2055-56  Kharif  109000  171500  1.57
238  JALNAR  Bajra  2055-56  Kharif  35000  49000  1.40
239  NASHIK  Bajra  2055-56  Kharif  171200  230300  1.35
240  ANURAGCHAD  Bajra  2056-57  Kharif  89800  120500  1.34
241  BEED  Bajra  2056-57  Kharif  122000  112000  0.92
242  DARE  Bajra  2056-57  Kharif  109000  171500  1.57
243  JALNAR  Bajra  2056-57  Kharif  35000  49000  1.40
244  NASHIK  Bajra  2056-57  Kharif  171200  230300  1.35
245  ANURAGCHAD  Bajra  2057-58  Kharif  89800  12
```

5.2 Output Report.

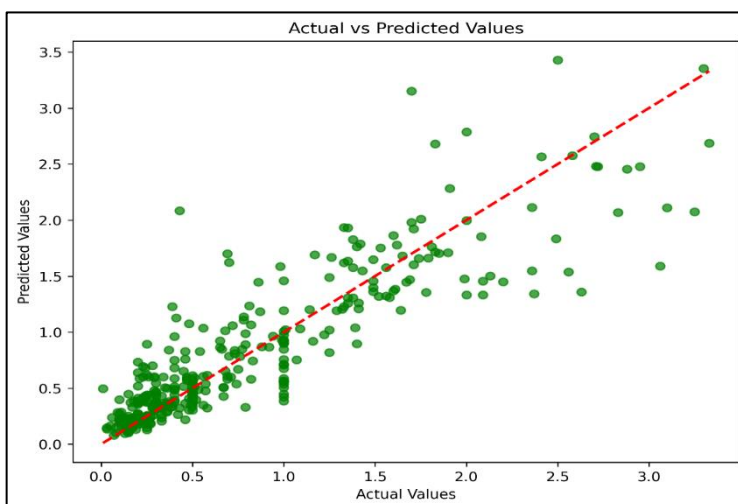
1] from Python:

1. Dataset without any Filtration



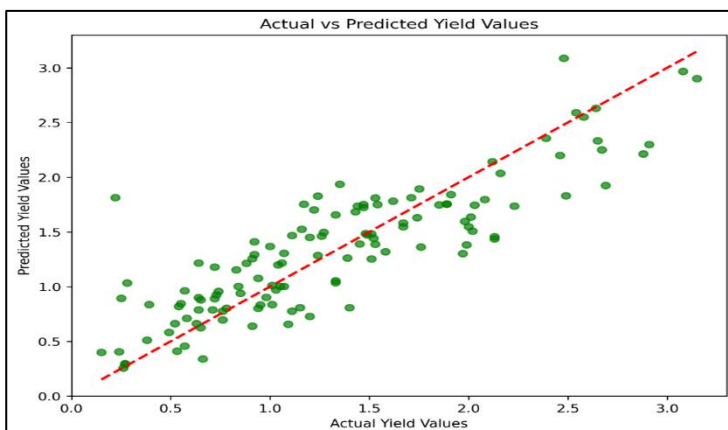
1. Here, I have kept the dataset as it is (not changed or filtered data).
2. I have not removed or capped outliers here.
3. I have 2 crops which have the production higher than other crops (that's why I got the final plot like this), the crops are sugarcane and maize.
4. Also, this dataset has given me the highest accuracy for my prediction i.e. (97.84 %).

2. Filtered Dataset with Outliers



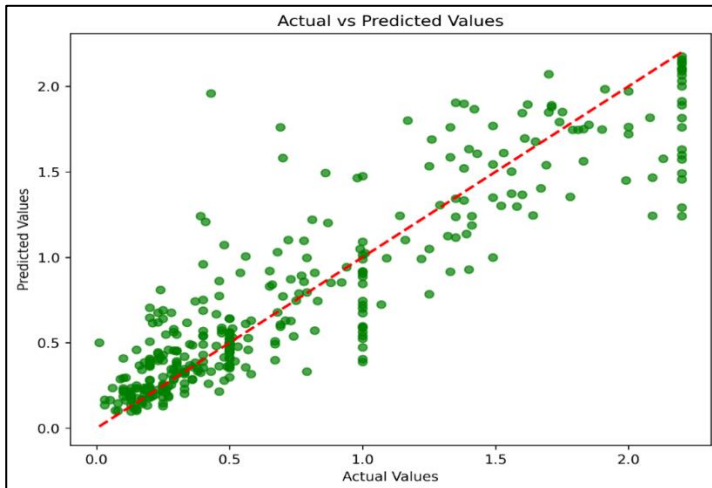
1. Here, I actually have removed all the sugarcane records but kept the outliers.
2. As the records for sugarcane for my dataset were very less (78 out of 2110).
3. Also, I have only kept the records of Kharif and Rabi Season (removed the summer and Whole year records).
4. This model gave me the somewhat of better axis at last for my prediction plot i.e. approx. to my actual yield values.
5. The Dataset gave me the accuracy of 78.71 %.

3. Filtered Dataset with Outliers Removed



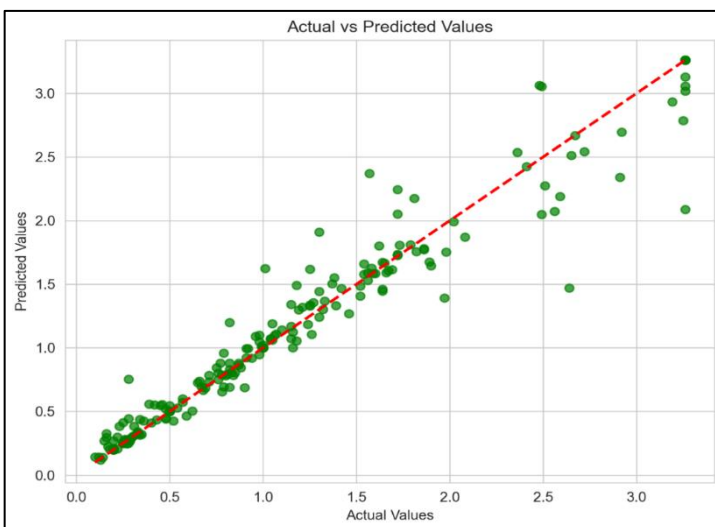
1. Here, I have used same data from Second Scenario (kept only Kharif and Rabi Crops).
2. But the only change is I removed the outliers present in this dataset.
3. For this data I am getting the plot best fit but the accuracy is not as expected.
4. These changes gave me the accuracy of 74.61 %.

4. Filtered Dataset with Capped Outliers



1. Here, I have used same data (Kharif and Rabi only).
2. Only change is I have capped the outliers (using lower and upper bound value).
3. I replaced the outliers with upper value which were greater than upper bound and same for lower bound also.
4. It resulted in improved accuracy than the removed outlier's case, but not upto the mark.
5. These changes gave me the accuracy of 81.04 %.

5. Final Dataset with Specific Changes



1. Here, I actually made some conditional changes (removed the values from area and production where values are less than 50) and from target column (Yield), I removed the values that were less than 0.5.
2. I did this to check if it has any effect on the accuracy of my models.
3. Also, I have capped the outliers here and the accuracy I got was the best so far rather than 1st Scenario.
4. The Accuracy these changes gave me is 95.54 %.

2) Power BI:

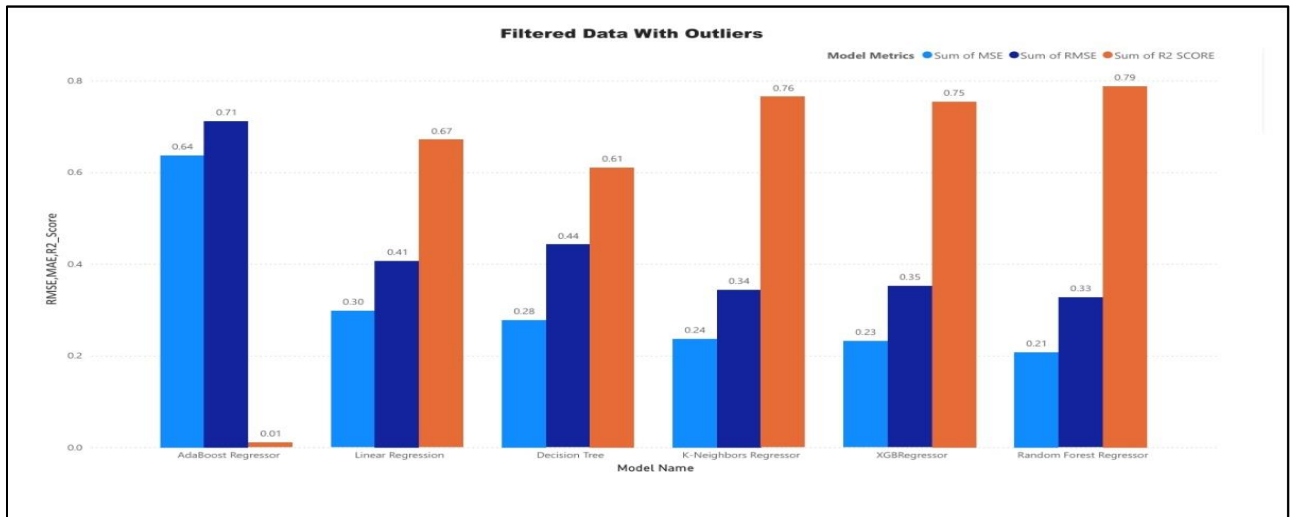


Fig – 1 : Filtered Data with Outliers.

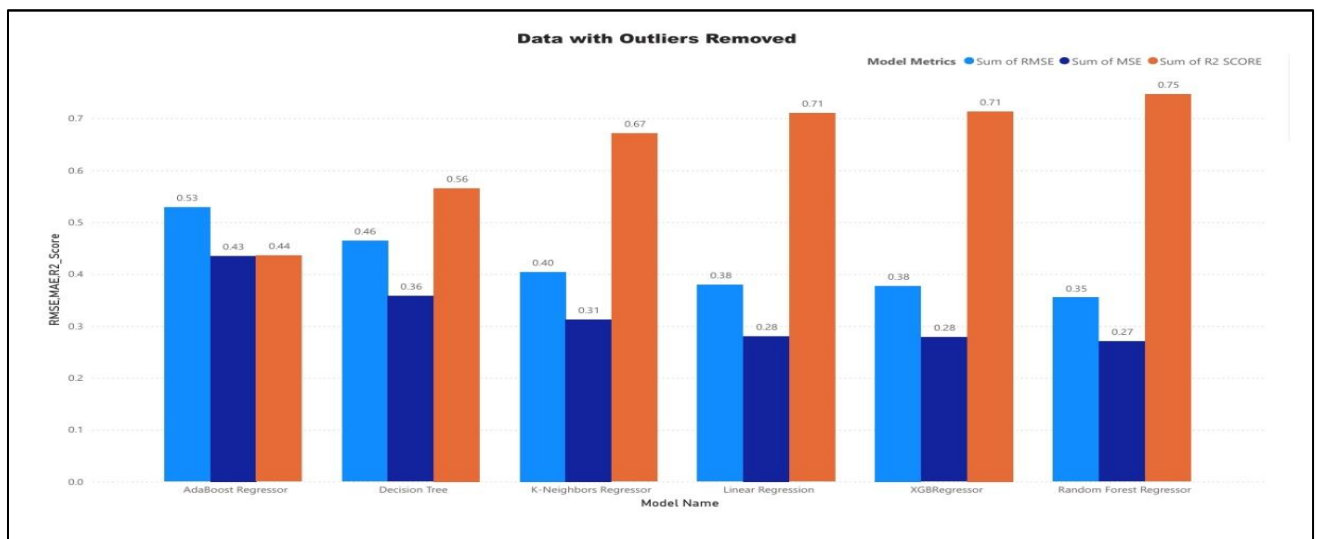


Fig - 2: Filtered Data with Outliers Removed.

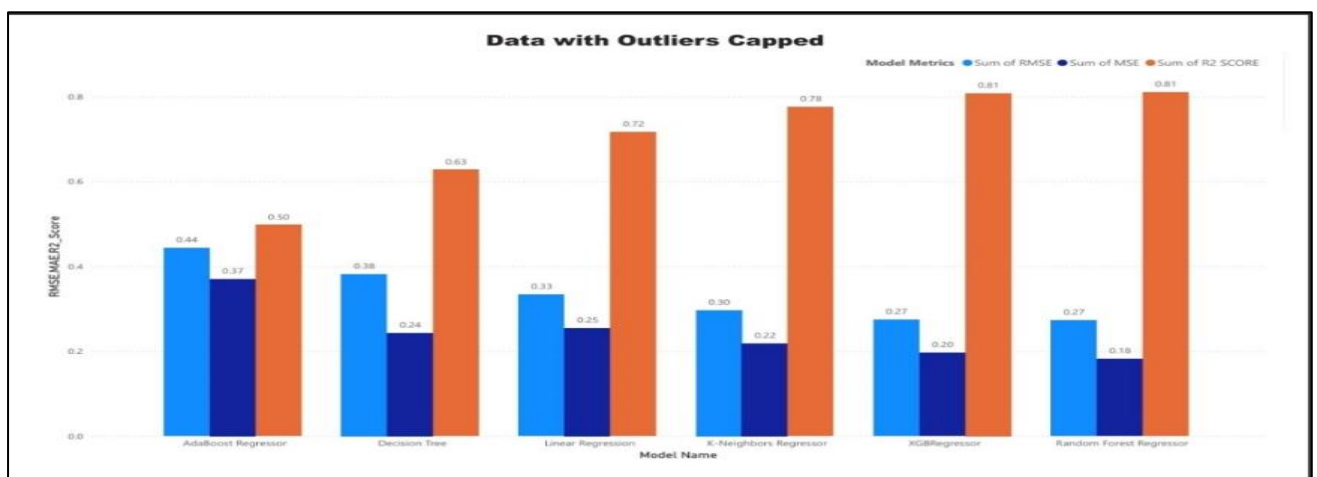


Fig -3: Filtered Data with Outliers Capped.

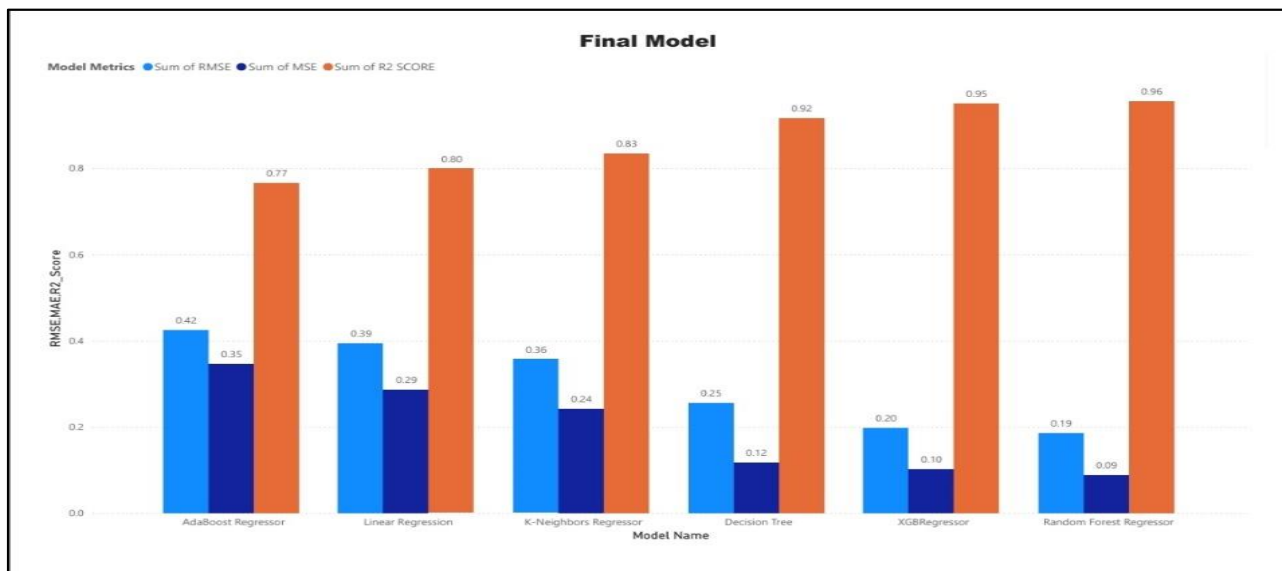


Fig-4: Final Model with specific changes.

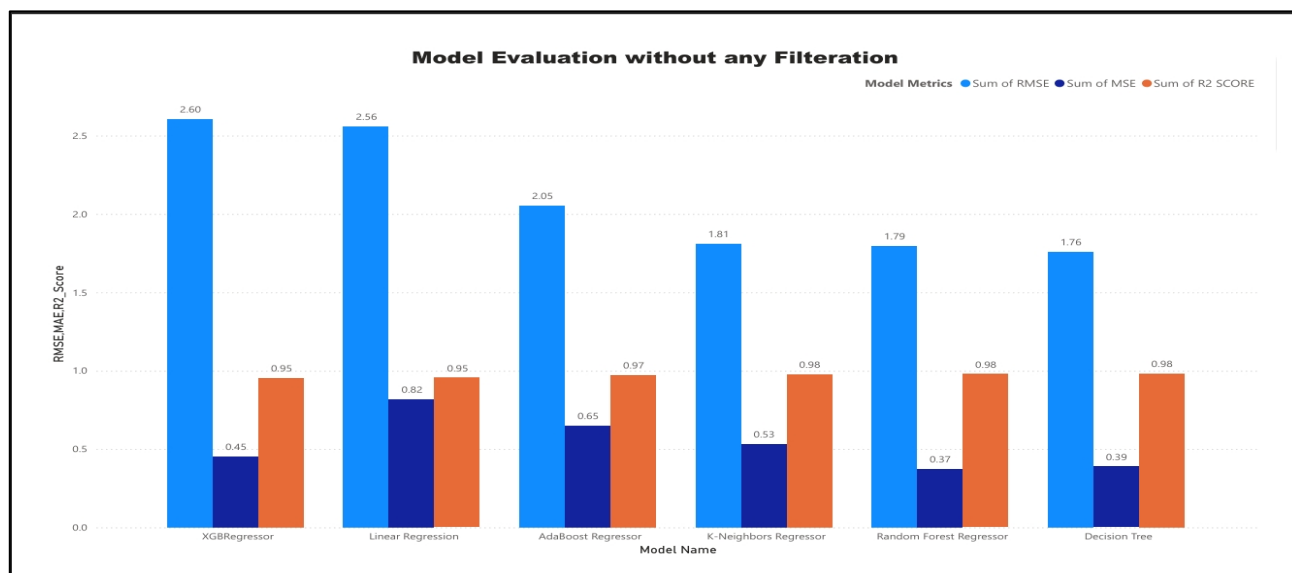


Fig-5: Data without any Filtration.

6. Future Enhancements

1. Integration of Remote Sensing Data:

Incorporating satellite imagery and vegetation indices such as NDVI, EVI, and LAI can significantly enhance the prediction accuracy of crop yield models. These indices provide continuous, real-time insights into vegetation health, canopy coverage, and growth stages. By combining remote sensing data with traditional datasets, the model can better capture spatial variability and environmental influences across large agricultural areas.

2. Inclusion of Climate Change Scenarios:

Integrating projected climate variables such as rainfall shifts, temperature rise, and CO₂ concentration enables the model to account for future environmental uncertainties. This enhancement makes the system more robust and adaptive to changing climatic conditions. It ensures that yield predictions remain reliable and relevant even under long-term climate fluctuations and extreme weather events.

3. Hybrid and Deep Learning Models:

The use of hybrid deep learning architectures that combine CNN, LSTM, and Transformer models can improve the model's ability to learn both spatial and temporal dependencies. CNNs can process spatial features from satellite or soil data, while LSTMs and Transformers can handle time-based patterns in weather and crop growth. Such integration results in a more comprehensive and accurate prediction framework.

4. Explainable AI (XAI) Integration:

Incorporating explainable AI methods such as SHAP and LIME enhances the interpretability of the model by identifying which features contribute most to the prediction. This transparency is crucial for building trust among farmers, policymakers, and researchers. It allows stakeholders to make data-driven decisions with a clear understanding of how and why the model produces specific outcomes.

5. IoT and Sensor Data Utilization:

Integrating Internet of Things (IoT) devices such as soil moisture sensors, temperature probes, and local weather stations provides continuous, real-time field data. This information can dynamically update yield predictions and detect changes in soil or crop conditions. The combination of IoT and ML supports precision agriculture by enabling timely interventions and improving resource efficiency.

6. Decision Support System (DSS) Development:

Developing a user-friendly Decision Support System in the form of a web or mobile application can make the model easily accessible to farmers and agricultural planners. Such a system can provide yield forecasts, crop management suggestions, and early warnings based on predictive analytics. This enhancement bridges the gap between research and practical implementation, promoting data-driven farming decisions.

7. Conclusion

- a) This project demonstrated the effectiveness of machine learning models in predicting crop yield, that can help the farmers to improve the productivity more efficiently.
- b) The results highlight the importance of features like season and area that has a very strong effect on increasing the yield.
- c) While models like Decision Tree can perform well under clean datasets, ensemble methods such as Random Forest offer more robust and generalizable outcomes.
- d) Overall, the study shows that machine learning can serve as a valuable tool in agricultural decision-making, helping farmers and policymakers plan more effectively for improved productivity.

8. Bibliography

1. Kaggle. *Agri Dataset*. Retrieved from <https://www.kaggle.com>
2. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). *Scikit-learn: Machine Learning in Python*. *Journal of Machine Learning Research*, 12, 2825-2830.
3. Microsoft. *Power BI Documentation*. Retrieved from <https://powerbi.microsoft.com>
4. Subramaniam, L. K., & Marimuthu, R. (2024). Crop Yield Prediction Using Deep Learning. *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, 8, 100611.
5. Surana, R., & Khandelwal, R. (2024). Crop Yield Prediction Using Machine Learning: A Pragmatic Approach.
6. Bhat, S. A., Qadri, S. A. A., Dubbey, V., Sofi, I. B., & Huang, N.-F. (2024). Impact of crop management practices on maize yield: Insights from farming in tropical regions and predictive modeling using machine learning. *Journal of Agriculture and Food Research*, 18, 101392. <https://doi.org/10.1016/j.jafr.2024.101392>
7. Pravesh, S. Y., Garg, N., Arora, R., Singh, S., & Sankari, S. S. (2024). Predictive modeling of crop yield using deep learning-based Transformer with climate change effects. *International Research Journal of Multidisciplinary Technovation*, 6(6), 223–240. <https://doi.org/10.54392/irjmt24616>
8. Javed, M. A., & Murad, M. A. A. (2024). Crop yield prediction in agriculture: A comprehensive review of machine learning and deep learning approaches, with insights for future research and sustainability. *Heliyon*, 10, e40836. <https://doi.org/10.1016/j.heliyon.2024.e40836>
9. Shawon, S. M. M., Ema, F. B., Mahi, A. K., Niha, F. L., & Zubair, H. T. (2025). Crop yield prediction using machine learning: An extensive and systematic literature review. *Smart Agricultural Technology*, 10, 100718. <https://doi.org/10.1016/j.atech.2024.100718>
10. Jhajharia, K., Mathur, P., Jain, S., & Nijhawan, S. (2023). Crop yield prediction using machine learning and deep learning techniques. *Procedia Computer Science*, 218, 406–417. <https://doi.org/10.1016/j.procs.2023.01.023>
11. Panigrahi, B., Kathal, K. C. R., & Sujatha, M. (2023). A machine learning-based comparative approach to predict the crop yield using supervised learning with regression models. *Procedia Computer Science*, 218, 2684–2693. <https://doi.org/10.1016/j.procs.2023.01.241>

12. Joshua, S. V., Priyadharson, A. S. M., Kannadasan, R., Khan, A. A., Lawanont, W., Khan, F. A., Rehman, A. U., & Ali, M. J. (2022). Crop yield prediction using machine learning approaches on a wide spectrum. *Computers, Materials & Continua*, 72(3), 5663–5678. <https://doi.org/10.32604/cmc.2022.027178>
13. Rajakumaran, R., Anandan, P., & Jeyaprakash, B. (2024). A hybrid model of crop yield prediction using support vector machine with principal component analysis and genetic algorithm. *Measurement: Sensors*, 33, 101002. <https://doi.org/10.1016/j.measen.2023.101002>